

Report Excited-State Methods

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The program files involved for question 1 and 2 are:

- AO_to_MO.f90 (Exercise 1)
- CIS.f90 (Exercise 2)
- TDHF.f90 (Exercise 3, 4)
- EST.f90

Those files can be found in the EST/src directory.

1. Configuration Interaction with Singles

Exercise 1:

Initially the four-index transformation is done in the 8 loops for each coefficient, and it was time consuming because it scales as N^8 . After converting it to the more efficient transform index after index, the scaling is reduced to N^5 .

Exercise 2:

The number of virtual orbital is equal to number of basis set minus the number of occupied orbitals. The dimension of A (OV) is then number of occupied orbitals times the number of virtual orbitals. Then the orbital energy difference, is corresponds to the diagonal with dimension OV. Then we can calculate the matrix A.

The values here are taken for the lowest energy difference corresponding to the energy difference between the ground state and the first singlet excited state. The value in the “()” parentheses refer to the CIS calculation made in the ORCA quantum chemistry software package. From this we can compare the value obtained in this exercise, and we can see that the values are in good agreement. This show that the calculation is correct.

Atom / basis set	Pvdz (a.u)	Pvtz (a.u)	Pvqz (a.u)
Helium	1.9090067538565250 (1.909008)	1.1688923065438752 (1.206076)	1.0349500856976275 (1.083669)
Beryllium	0.19457334057734402 (0.194598)	0.18900795271098683 (0.189196)	-
Neon	1.8010801896241528 (1.801028)	1.2204514651327483 (1.357903)	-

The CIS method is usually overestimated, because it is a poor estimate of the true excitation energies. Since we use the orbital energy difference $\epsilon_a - \epsilon_i$, is not related to the excitation energy if we use the canonical Hartree-Fock orbitals. Furthermore, the correlation is neglected within the CIS method., and we will expect an order of 1eV errors on the electron pairs.

2. Time-Dependent Hartree-Fock

Table below recorded the values of the TDHF excitation energies. We can see that the values are similar to CIS. Unfortunately, ORCA does not implement TDHF calculation to let us verify the result.

Atom / basis set	Pvdz (a.u)	Pvtz (a.u)	Pvqz (a.u)
Helium	1.89540186071867	1.1644562684436	1.0329429203633
Beryllium	0.188744661557623	0.186550324777951	-
Neon	1.80948259625311	1.22588303785787	-

Comparing CIS and TDHF, we can see that TDHF contains the eigenvalues from the CIS. Therefore, when the matrix B is equal to zero, the TDHF reduces to CIS. This is because TDHF is an extension of the CIS, and it does not only contains the singly excited states, but also the singly de-excited states. We can expect that TDHF is an improved transition moments compared to CIS. However, TDHF is still overestimate the energies. These methods perform well for low-lying electronic states that are not charge transfer transitions.

CIS-He cc-pvdz	TDHF-He cc-pvdz	CIS-He cc-pvtz	TDHF-He cc-pvtz
1.909006754	1.895401861	1.168892307	1.164456268
2.843557661	2.837653348	1.890125592	1.885645654
2.843557661	2.837653348	1.890125592	1.885645654
2.843557661	2.837653348	1.890125592	1.885645654
		3.828441329	3.826436169
		6.461850923	6.461358207
		6.461850923	6.461358207
		6.461850923	6.461358207
		6.461850923	6.461358207
		6.461850923	6.461358207
		6.461850923	6.461358207
		7.919587045	7.918072384
		7.919587045	7.918072384
		7.919587045	7.918072384
		8.462698884	8.46025313

The table shows the examples of the excitation energies obtained with pvdz and pvtz for He atom. We can see that the values are close, as they have similar properties. We can also see that the excitation energies calculated with TDHF, are slightly smaller than the CIS. Therefore, the effect of B matrix is small, and with the consideration of cheaper computation power of CIS, TDHF calculation does not significantly improve compared to CIS.

The full result for He, Be and Ne for each basis set is recorded in the result.xlsx file.

References:

- 1. Single-Reference ab Initio Methods for the Calculation of Excited States of Large Molecules.** Andreas Dreuw and Martin Head-Gordon. *Chemical Reviews* **2005** 105 (11), 4009-4037. DOI: 10.1021/cr0505627